

Jason Liu

☎ 647-721-9986 | ✉ jzliu@uwaterloo.ca | 💼 Jason Liu | 🌐 jasonliu1103 | 🌐 Portfolio Website

Education

University of Waterloo

Waterloo, ON

Candidate for Bachelor of Applied Science in Computer Engineering, Honours, Co-op

Sept. 2022 - Apr. 2027

Relevant Courses: Computer Networks, Computer Security, Systems Programming & Concurrency, Database Systems, Software Design & Architecture, Algorithms & Data Structures

Technical Skills

Languages: Python, Go, C#, TypeScript, Java, C/C++, SQL

Frameworks & Libraries: .NET/ASP.NET Core, React.js, Next.js, Node.js, Prisma, Tailwind CSS, Playwright

Tools: Git, CI/CD, PostgreSQL, Microsoft SQL Server, Terraform, GCP, OpenAI API

Experience

Software Engineer & Consultant Intern

Sept. 2026 - Present

IBM

Markham, ON

- Building **Python** automation for internal infrastructure workflows on a **mainframe DevOps** team, replacing manual configuration and validation steps across staging and production.
- Evaluating **LLM-assisted tooling** for infrastructure troubleshooting, including prompt design and failure-mode analysis for operational runbooks.

Software Engineer Intern

Jan. 2026 - Apr. 2026

Marketing Kitchen

Markham, ON

- Removed a hard **ERP dependency** from downstream services with a **.NET Web API** in **C#** wrapping a legacy **SOAP** system behind **REST endpoints**, translating XML into a stable JSON contract.
- Built a **.NET BackgroundService** pipeline that incrementally syncs orders into a **4-level SQL Server hierarchy** every **15 minutes**, using **idempotent upserts** so overlapping runs converge to the same state.
- Refactored repetitive per-entity persistence into a single **generic base class**, so adding a synced entity required only a DTO mapping.
- Parallelized **300+** upstream API calls behind **semaphore-based rate control** and response caching, keeping bulk ingestion reliable without triggering upstream throttling.

Software Engineer Intern

Jan. 2025 - Apr. 2025

Ford Motor Company

Oakville, ON

- Automated dependency triage on the notifications platform by wiring **GitHub Dependabot** into **Jira**, routing update tickets across **10+ repositories** previously tracked by hand.
- Made an internal configuration editor **schema-driven** in **React.js**, so new field types could be added through configuration instead of front-end changes.
- Built functional test coverage for **email delivery endpoints**, catching endpoint regressions before release across staging and production deploys.
- Refactored **3 OpenTelemetry Terraform modules** into reusable components, cutting observability provisioning time by **25%** across environments.

Projects

Prediction Market - Ledger & Settlement | *Python, FastAPI, SQLite, Kotlin/Android*

Aug. 2026

- Owned the **money system** for a 5-person prediction market app, building an **append-only transaction ledger** with cached balances where every balance change funnels through a **single write path**, so balances can always be reconciled against the ledger.
- Implemented **five transaction types** (bet, payout, refund, reward, stipend) as independent components over that path, including **probability-scaled settlement** based on market odds at position entry.
- Modeled cancellations as **compensating refund entries** rather than deletions, preserving a full audit trail and keeping settlement reconstructable from the ledger.

Pulseboard 🎧 | *Next.js, TypeScript, PostgreSQL, Prisma, Spotify API*

Apr. 2026

- Built a **full-stack Spotify analytics platform** on a **Prisma-backed PostgreSQL** schema that snapshots listening history over time, enabling trend queries the Spotify API does not expose.
- Implemented a **scheduled sync pipeline** ingesting **150+ tracks and artists** every **30 minutes** as incremental snapshots, keeping trends accurate without full refetches.